Water Pollution

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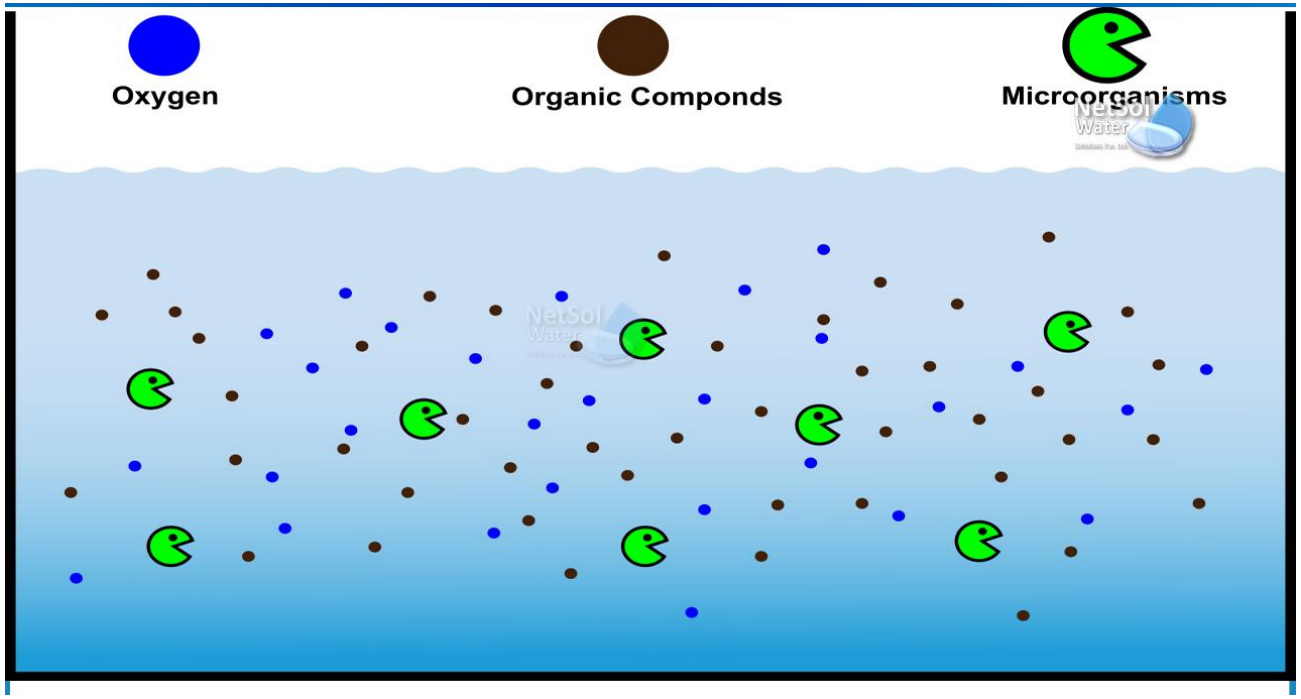
Biochemical Oxygen Demand (BOD)

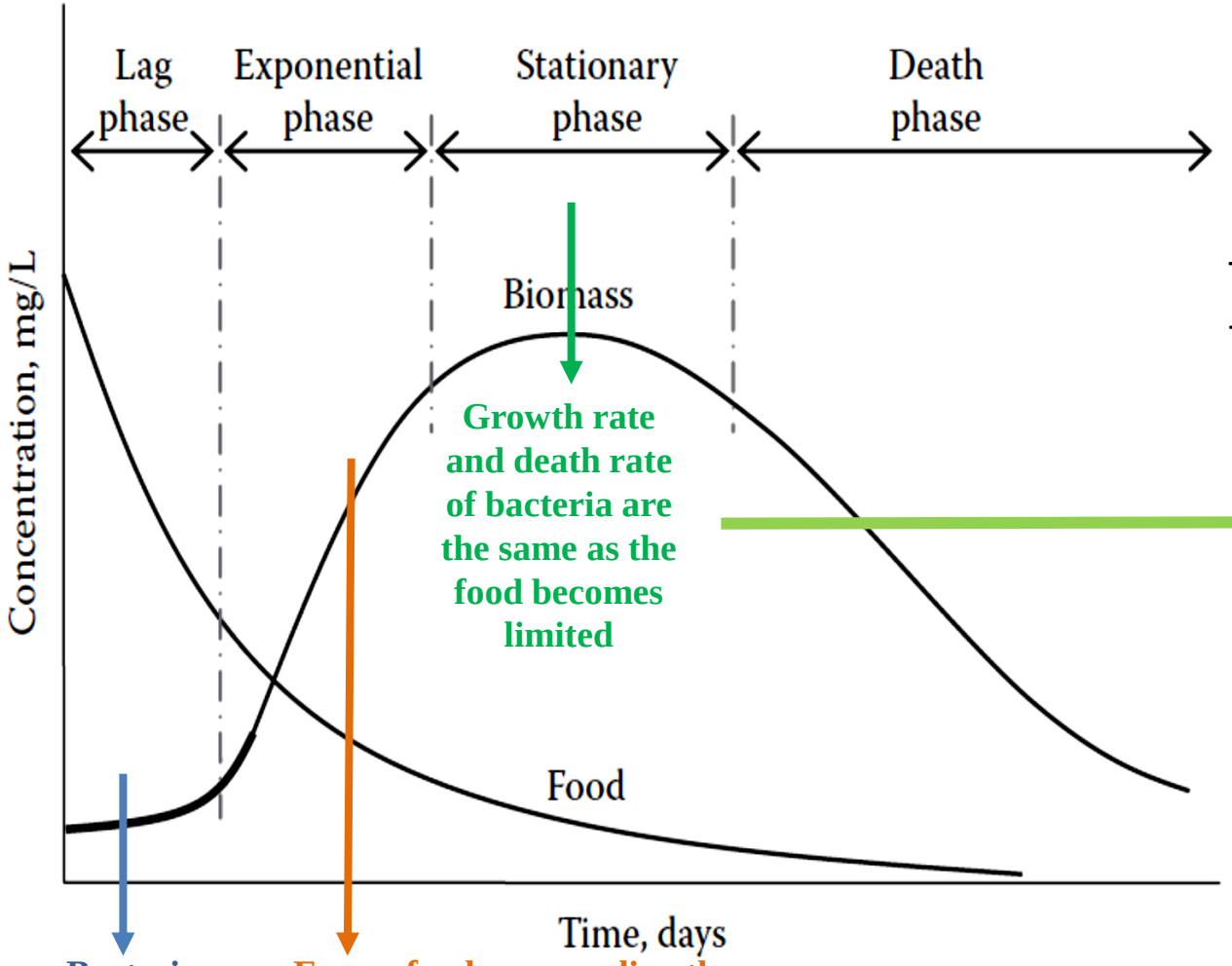
- Biodegradable materials consists of organics that can be utilized for food by naturally occurring microorganisms within a reasonable length of time.
- Biochemical oxygen demand (BOD) represents the amount of oxygen consumed by bacteria and other microorganisms while they decompose organic matter under aerobic (oxygen is present) conditions at a specified temperature.
- Intention: To measure the strength of organic matter in wastewater



$$C_XH_YO_ZN\ (\text{Organics}) + O_2 \xrightarrow{\textit{Microorganisms}} CO_2 + H_2O + \text{New Cells}$$

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: BOD, https://www.netsolwater.com/measurement-of-biological-oxygen-demand-or-bod.php?blog=1400 , Accessed on: 1st Feb, 2025.

Growth Curve of Microorganisms



Growth rate and death rate of bacteria are the same as the food becomes limited

Bacteria acclimate to the new environment

Excess food surrounding the bacteria; rate of metabolism and growth is a function of the ability of microorganism to process the substrate

The rate at which organics are utilized by micro organisms is a first-order reaction-

$$\frac{dX}{dt} = kX$$

Where,

– $\frac{dX}{dt}$ is the growth rate of the biomass (mass dry wt/vol.t)

–X is cell concentration (mass dry wt/vol)

– k is the specific growth rate (time-1)

– t is time

Numerical

E. coli cells take 2 hrs. produce 1024 cells. How much time they will take to produce 2048 cells? (Generation time of *E.coli* is 20 min)

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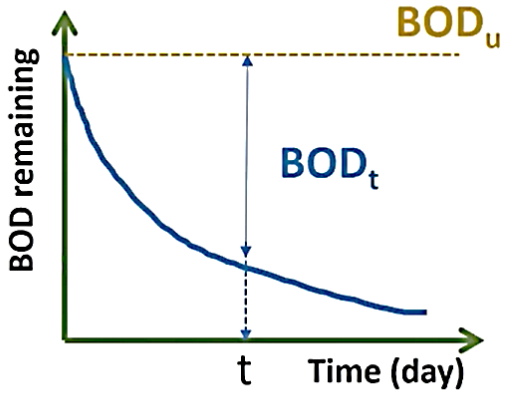
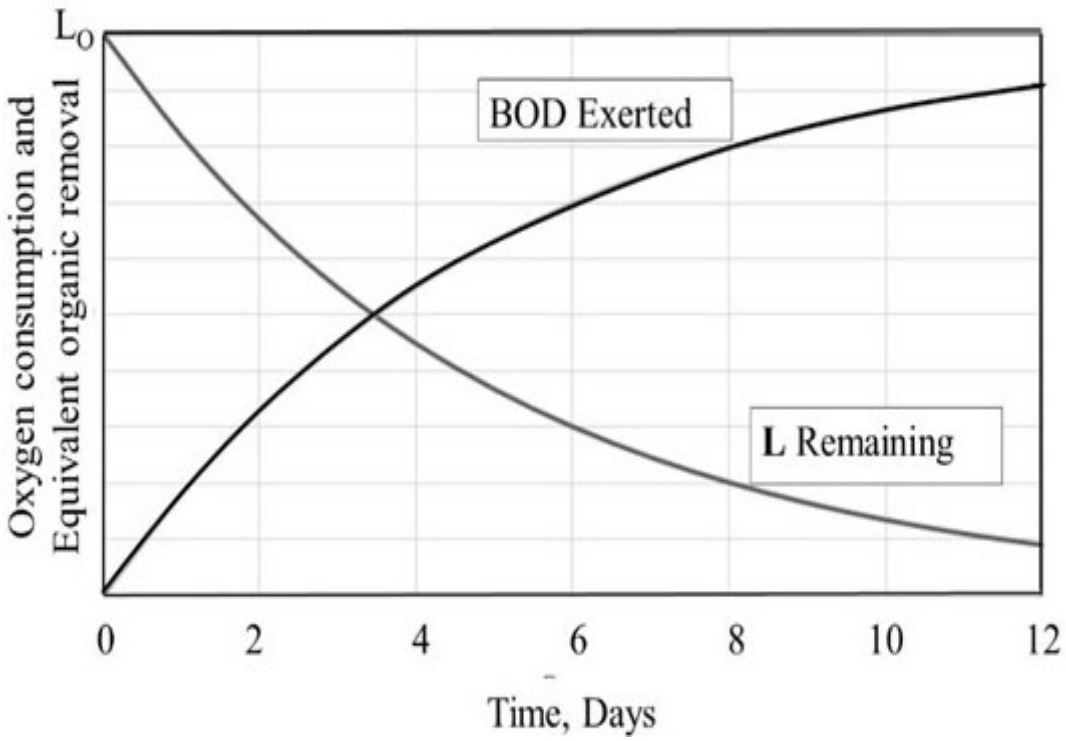


BOD$_u$ (ultimate BOD)

Amount of oxygen needed to decompose all the organic matter (mg O_2/L)

BOD$_t$

Oxygen consumed in water after t days (mg O_2/L)



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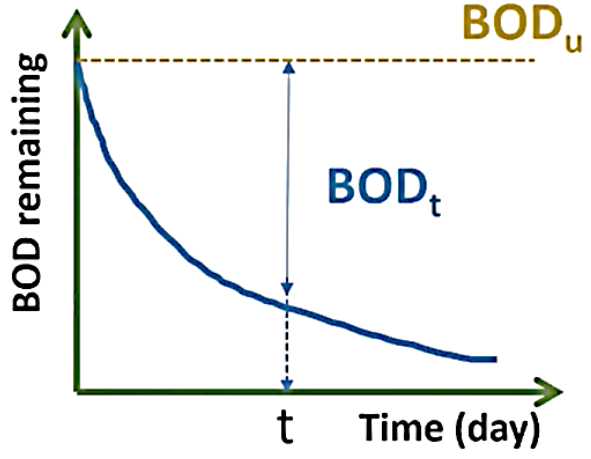


BOD$_u$ (ultimate BOD)

Amount of oxygen needed to decompose all the organic matter (mg O_2/L)

BOD$_t$

Oxygen consumed in water after t days (mg O_2/L)

$$k_T = k_{20}\theta^{T-20^\circ}$$

θ *is often assumed to be* 1.047 *although it is known to vary* somewhat with temperature.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: BOD, https://www.netsolwater.com/measurement-of-biological-oxygen-demand-or-bod.php?blog=1400 , Accessed on: 1st Feb, 2025.

BOD Measurement

For BOD, Saturation oxygen concentration of 9 mg/L in dilution water, with a minimum of 2 and a maximum of 7 mg/L of O_2 being consumed.

Titrant: Sodium Thiosulfate

BOD_5 is calculated by:

- Unseeded : $\mathrm{BOD}_5 = \dfrac{(D_0 - D_5)}{P}$
- Seeded: $\mathrm{BOD}_5 = \dfrac{(D_0 - D_5) - (B_0 - B_5)f}{P}$

where:

D_0 is the dissolved oxygen (DO) of the diluted solution after preparation (mg/L)

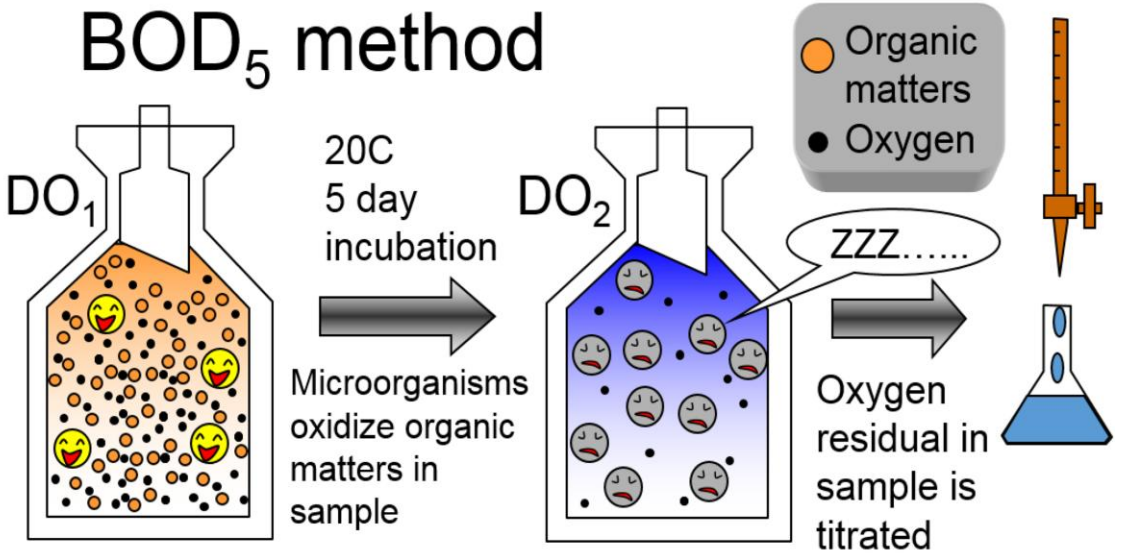
D_5 is the DO of the diluted solution after 5 day incubation (mg/L)

P is the decimal dilution factor

B_0 is the DO of diluted seed sample after preparation (mg/L)

B_5 is the DO of diluted seed sample after 5 day incubation (mg/L)

f is the ratio of seed volume in dilution solution to seed volume in BOD test on seed



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Arlyapov, V. A., Plekhanova, Y. V., Kamanina, O. A., Nakamura, H., & Reshetilov, A. N. (2022). Microbial biosensors for rapid determination of biochemical oxygen demand: approaches, tendencies and development prospects. Biosensors, 12(10), 842.

Numerical

The BOD of a wastewater is suspected to range from 50 to 200 mg/L. Three dilutions are prepared to cover this range. The procedure is the same in each case. First the sample is placed in the standard BOD bottle and is then diluted to 300 mL with organic-free, oxygen-saturated water. The initial dissolved oxygen is determined and the bottles tightly stoppered and placed in the incubator at 20°C for 5 days, after which the dissolved oxygen is again determined. Determine the BOD of the sample.

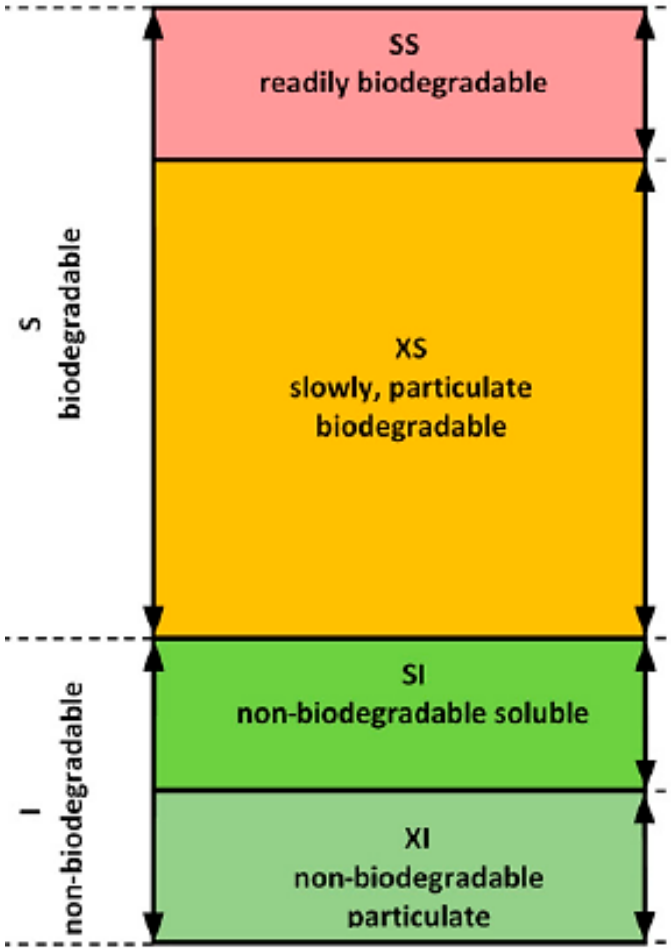
Wastewater, milliliter	DO_i, mg/L	DO_5, mg/L
5	9.2	6.9
10	9.1	4.4
20	8.9	1.5

Numerical

The BOD_5 of a wastewater is determined to be 150 mg/L at 20°C. The k value is known to be 0.23 per day. What would the BOD_8 be if the test were run at 15°C?

Chemical Oxygen Demand (COD)

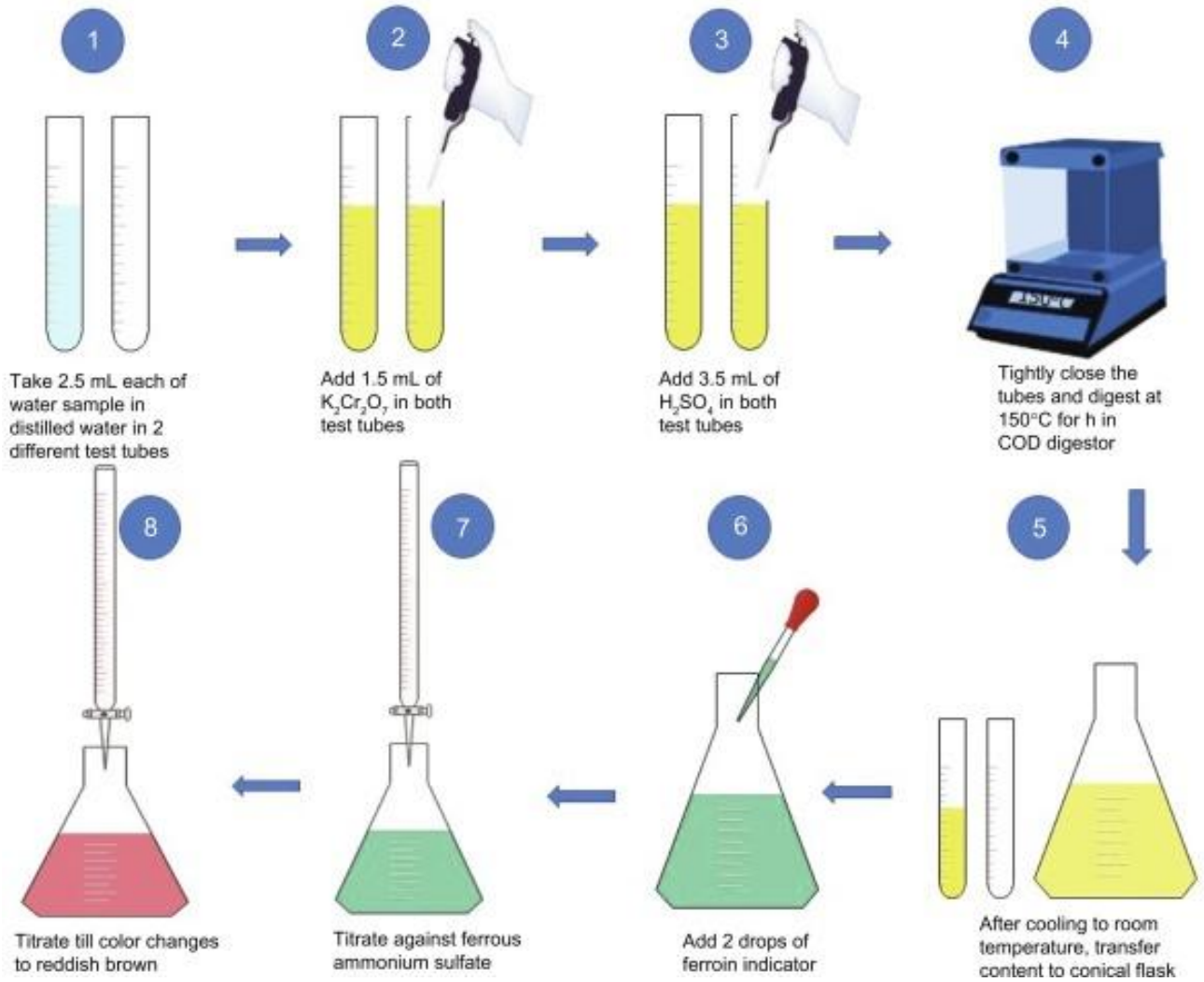
- Chemical oxygen demand (COD) represents the amount of oxygen required to oxidize the biodegradable as well as the non biodegradable organic fraction of the wastewater sample.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Płuciennik-Koropczuk, E., & Myszograj, S. (2019). New approach in COD fractionation methods. Water, 11(7), 1484.

Chemical Oxygen Demand (COD)

- Add 2.5 mL of diluted sample into the digestion vessel, add 1.5 mL of digestion solution along with 3.5 mL of sulphuric acid reagent.
- Keep the digestion vessel into the reactor for 2 h (Ensure the reactor temp. 148˚C).
- After the completion of digestion, take out the digestion vessel, keep them for 20 min at room temp.
- Add 2 drops of ferroin indicator solution and titrate with 0.1 M Ferrous Ammonium Sulphate solution. The end point is a sharp colour change from blue green to reddish brown, although the blue green may reappear within minutes.



Source: 1. Singh, S., Singh, J., & Singh, H. (2021). Chemical oxygen demand and biochemical oxygen demand. In Green Sustainable Process for Chemical and Environmental Engineering and Science (pp. 69-83). Elsevier.

Chemical Oxygen Demand (COD)

- **Step 1: Oxidation of organic matter and Reduction of Potassium Dichromate**

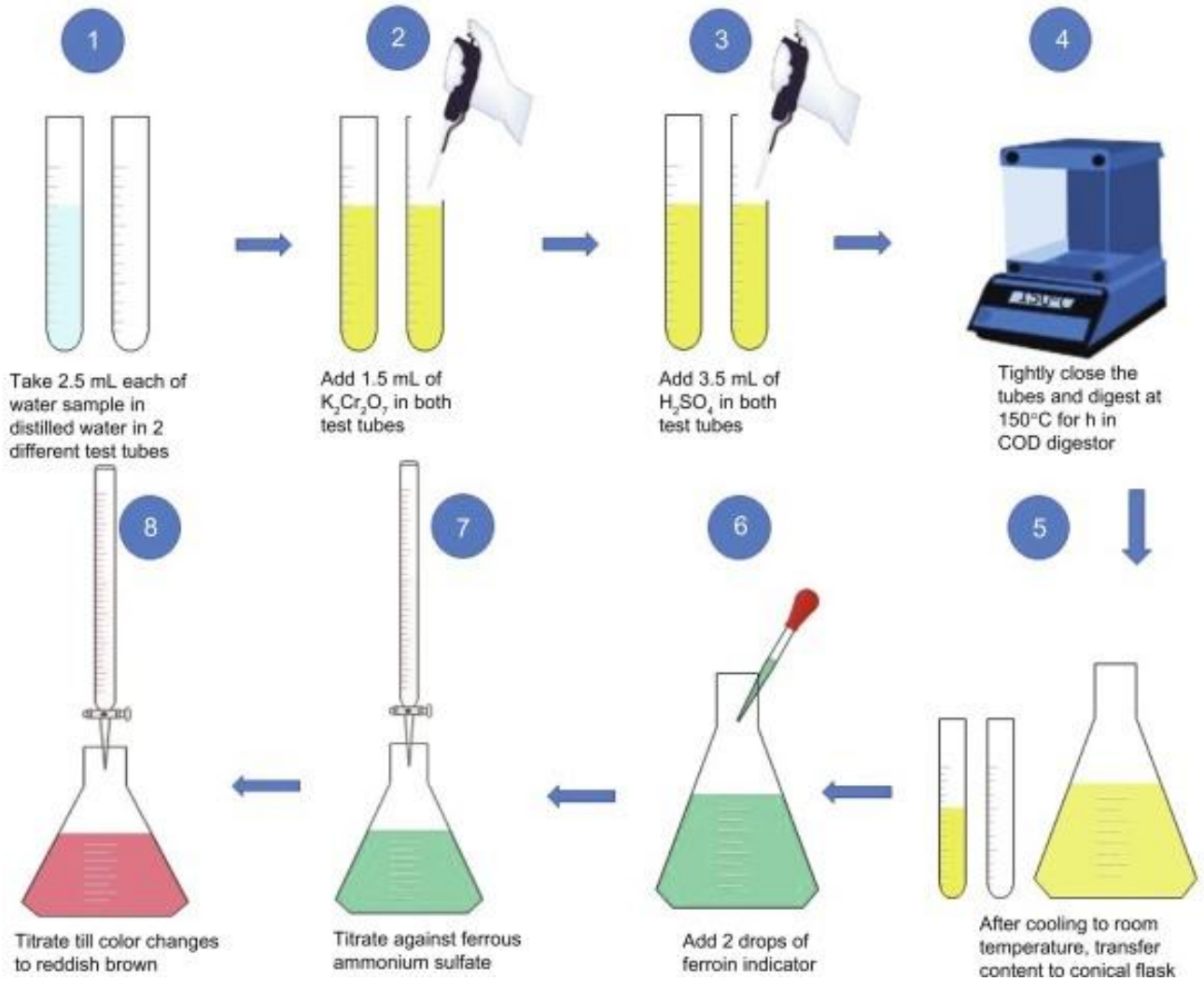
$$\text{Organic Matter} + K_2Cr_2O_7 + H_2SO_4 \longrightarrow \text{Oxidized Products} + Cr^{3+} + H_2O$$

$$\text{R-H} + O_2 \longrightarrow CO_2 + H_2O$$

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

- Step 3: Titration with FAS

$$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$



Source: 1. Singh, S., Singh, J., & Singh, H. (2021). Chemical oxygen demand and biochemical oxygen demand. In Green Sustainable Process for Chemical and Environmental Engineering and Science (pp. 69-83). Elsevier.

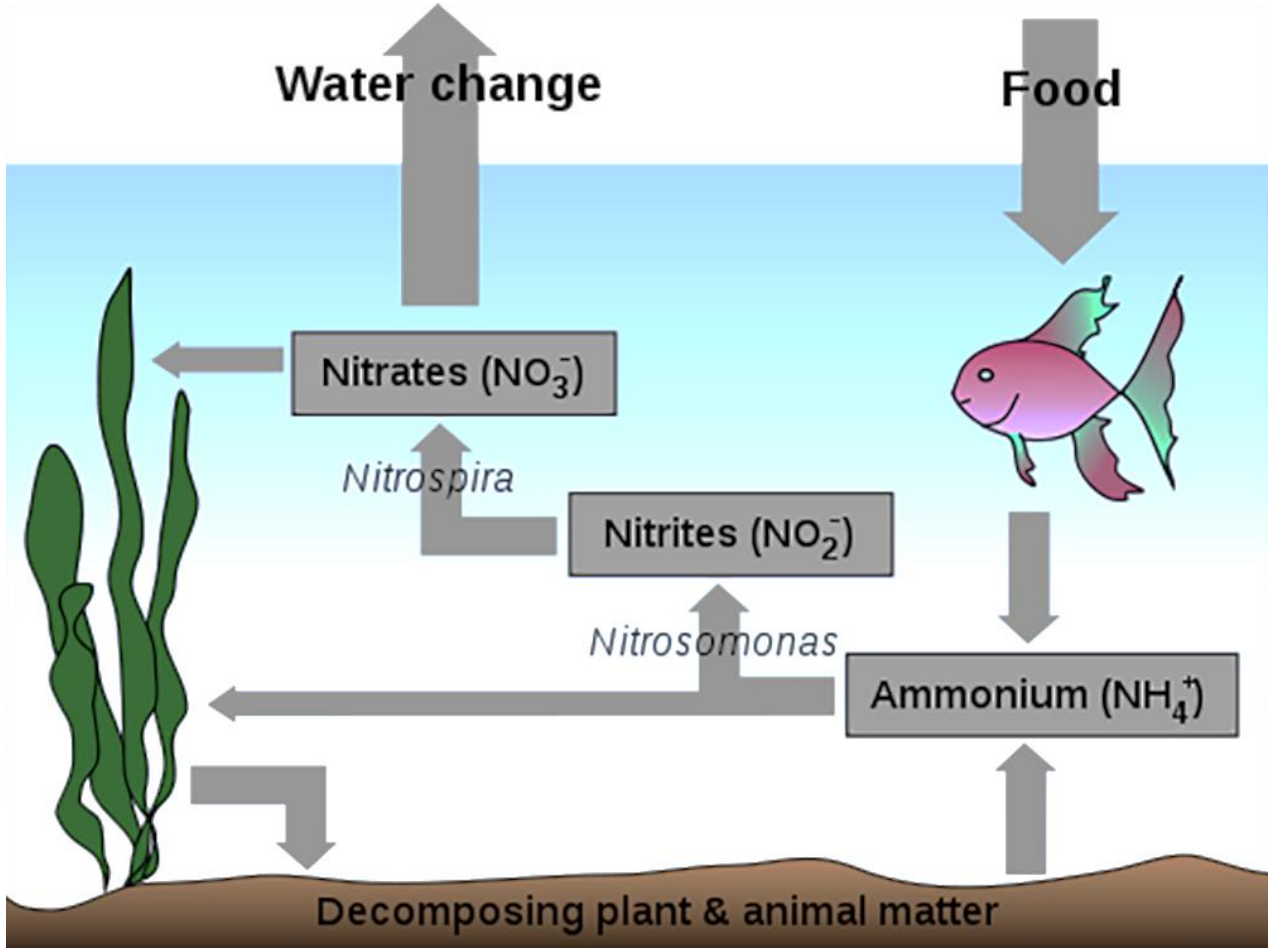
Numerical

What is the theoretical oxygen demand of 300 mg/L of glucose solution?

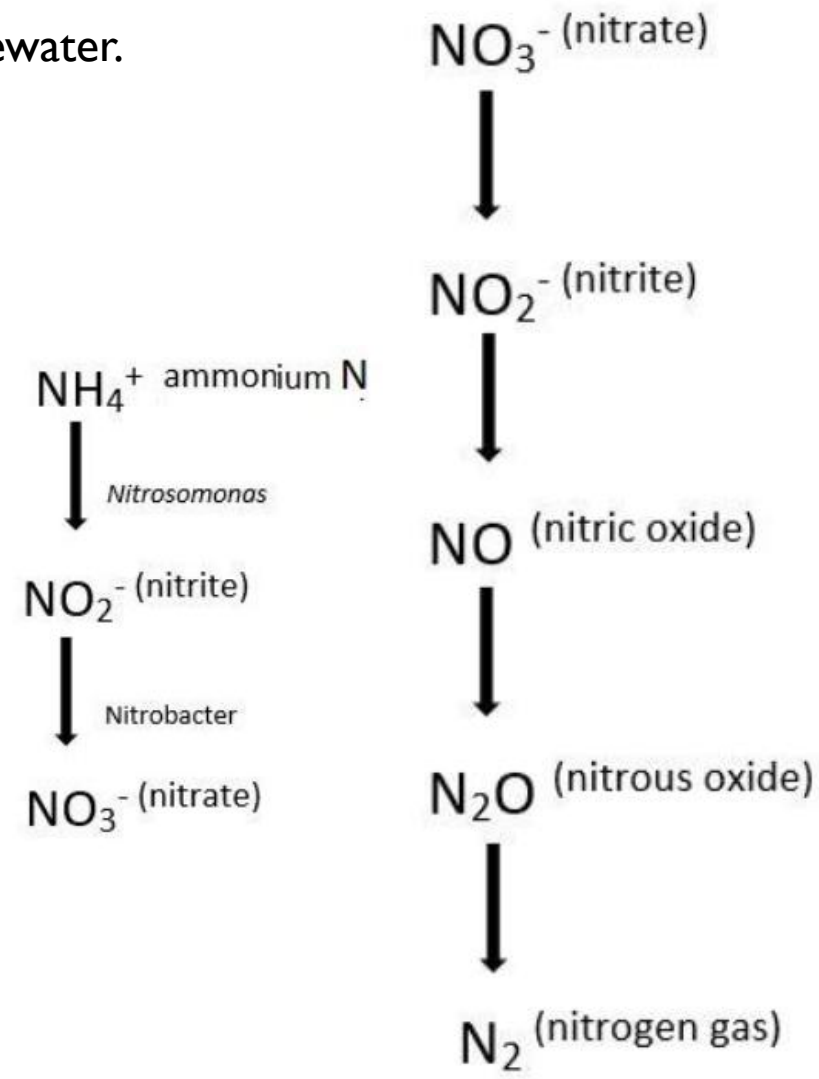
$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

Nitrogen

Ammonia is generally considered an indicator of recent pollution in wastewater.



Nitrification



Nitrification De-Nitrification

Source: 1. 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Nitrogen, https://water.mecc.edu/courses/Env211/lesson21.htm

Nitrogen

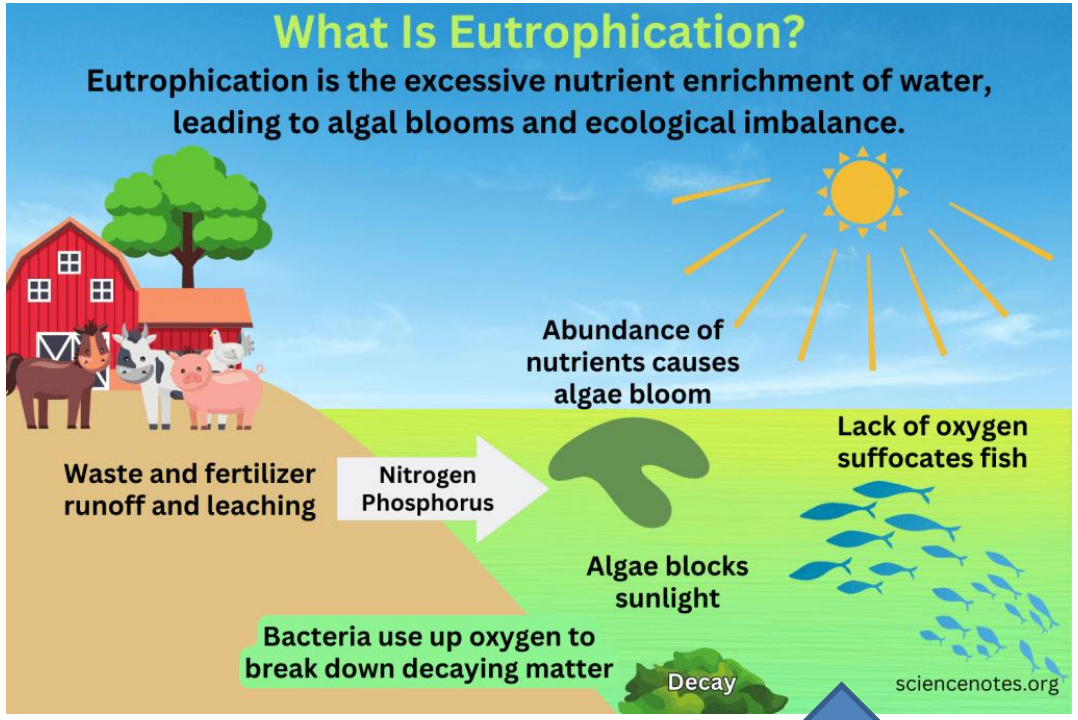
Basis for Comparison	Nitrification	Denitrification
Meaning	The part of nitrogen cycle where ammonium is converted into nitrate.	The level where reduction of nitrate is made into nitrogen gas.
The process involves	Nitrifying bacteria like *Nitrobacter*, *Nitrosomonas* Grows slowly Requires aerobic conditions	Denitrifying bacteria like *Spirillum*, *Lactobacillus*, *Pseudomonas*, *Thiobacillus* Grows rapidly Requires anaerobic conditions
The microbes	Autotrophic	Heterotrophic
Precursor	Ammonium	Nitrate
End product	Nitrate	Nitrogen
pH and Temperature	pH 6.5 - 8.5 Temperature: 16 - 35°C	pH 7.0 - 8.5 Temperature: 26 - 38°C
Importance	Provides nitrate to the plant, which acts as teh important nitrogen source	Is used in wastewater treatment and is beneficial for aquatic habitats.

Nitrate poisoning in infant animals, including humans, can cause serious problems and even death. Apparently, the lower acidity in an infant's intestinal tract permits growth of nitrate-reducing bacteria that convert the nitrate to nitrite, which is then absorbed into the bloodstream. Nitrite has a greater affinity for hemoglobin than does oxygen and thus replaces oxygen in the blood. complex. The body is denied essential oxygen and, in extreme cases, the victim suffocates Because oxygen starvation results in a bluish discoloration of the body, nitrate poisoning has been referred to as the **"blue baby" syndrome**, although the correct term is **methemoglobinemia**.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Nitrogen, https://water.mecc.edu/courses/Env211/lesson21.htm

Phosphorus

- While phosphates are not toxic and do not represent a direct health threat to human or other organisms, they do represent a serious indirect threat to water quality.
- When the available supply is increased, rapid growth of aquatic plants usual results, with severe consequences (Eutrophication).



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Phosphorus, https://sciencenotes.org/what-is-eutrophication-definition-and-explanation/#google_vignette

Drinking water standards as per **IS 10500 : 2012**

Sl No.	Characteristic	Requirement (Acceptable Limit)	Permissible Limit in the Absence of Alternate Source	Method of Test, Ref to Part of IS 3025	Remarks
(1)	(2)	(3)	(4)	(5)	(6)
i)	Colour, Hazen units, *Max*	5	15	Part 4	Extended to 15 only, if toxic substances are not suspected in absence of alternate sources
ii)	Odour	Agreeable	Agreeable	Part 5	a) Test cold and when heated b) Test at several dilutions
iii)	*p*H value	6.5-8.5	No relaxation	Part 11	—
iv)	Taste	Agreeable	Agreeable	Parts 7 and 8	Test to be conducted only after safety has been established
v)	Turbidity, NTU, *Max*	1	5	Part 10	—
vi)	Total dissolved solids, mg/l, *Max*	500	2 000	Part 16	—

NOTE — It is recommended that the acceptable limit is to be implemented. Values in excess of those mentioned under 'acceptable' render the water not suitable, but still may be tolerated in the absence of an alternative source but up to the limits indicated under 'permissible limit in the absence of alternate source' in col 4, above which the sources will have to be rejected.

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Sl No.	Characteristic	Requirement (Acceptable Limit)	Permissible Limit in the Absence of Alternate Source	Method of Test, Ref to	Remarks
(1)	(2)	(3)	(4)	(5)	(6)
i)	Aluminium (as Al), mg/l, *Max*	0.03	0.2	IS 3025 (Part 55)	—
ii)	Ammonia (as total ammonia-N), mg/l, *Max*	0.5	No relaxation	IS 3025 (Part 34)	—
iii)	Anionic detergents (as MBAS) mg/l, *Max*	0.2	1.0	Annex K of IS 13428	—
iv)	Barium (as Ba), mg/l, *Max*	0.7	No relaxation	Annex F of IS 13428* or IS 15302	—
v)	Boron (as B), mg/l, *Max*	0.5	1.0	IS 3025 (Part 57)	—
vi)	Calcium (as Ca), mg/l, *Max*	75	200	IS 3025 (Part 40)	—
vii)	Chloramines (as Cl_2), mg/l, *Max*	4.0	No relaxation	IS 3025 (Part 26)* or APHA 4500-Cl G	—
viii)	Chloride (as Cl), mg/l, *Max*	250	1 000	IS 3025 (Part 32)	—
ix)	Copper (as Cu), mg/l, *Max*	0.05	1.5	IS 3025 (Part 42)	—
x)	Fluoride (as F) mg/l, *Max*	1.0	1.5	IS 3025 (Part 60)	—
xi)	Free residual chlorine, mg/l, *Min*	0.2	1	IS 3025 (Part 26)	To be applicable only when water is chlorinated. Tested at consumer end. When protection against viral infection is required, it should be minimum 0.5 mg/l
xii)	Iron (as Fe), mg/l, *Max*	0.3	No relaxation	IS 3025 (Part 53)	Total concentration of manganese (as Mn) and iron (as Fe) shall not exceed 0.3 mg/l
xiii)	Magnesium (as Mg), mg/l, *Max*	30	100	IS 3025 (Part 46)	—
xiv)	Manganese (as Mn), mg/l, *Max*	0.1	0.3	IS 3025 (Part 59)	Total concentration of manganese (as Mn) and iron (as Fe) shall not exceed 0.3 mg/l
xv)	Mineral oil, mg/l, *Max*	0.5	No relaxation	Clause **6** of IS 3025 (Part 39) Infrared partition method	—
xvi)	Nitrate (as NO_3), mg/l, *Max*	45	No relaxation	IS 3025 (Part 34)	—
xvii)	Phenolic compounds (as C_6H_5OH), mg/l, *Max*	0.001	0.002	IS 3025 (Part 43)	—
xviii)	Selenium (as Se), mg/l, *Max*	0.01	No relaxation	IS 3025 (Part 56) or IS 15303*	—
xix)	Silver (as Ag), mg/l, *Max*	0.1	No relaxation	Annex J of IS 13428	—
xx)	Sulphate (as SO_4) mg/l, *Max*	200	400	IS 3025 (Part 24)	May be extended to 400 provided that Magnesium does not exceed 30
xxi)	Sulphide (as H_2S), mg/l, *Max*	0.05	No relaxation	IS 3025 (Part 29)	—
xxii)	Total alkalinity as calcium carbonate, mg/l, *Max*	200	600	IS 3025 (Part 23)	—
xxiii)	Total hardness (as $CaCO_3$), mg/l, *Max*	200	600	IS 3025 (Part 21)	—
xxiv)	Zinc (as Zn), mg/l, *Max*	5	15	IS 3025 (Part 49)	—

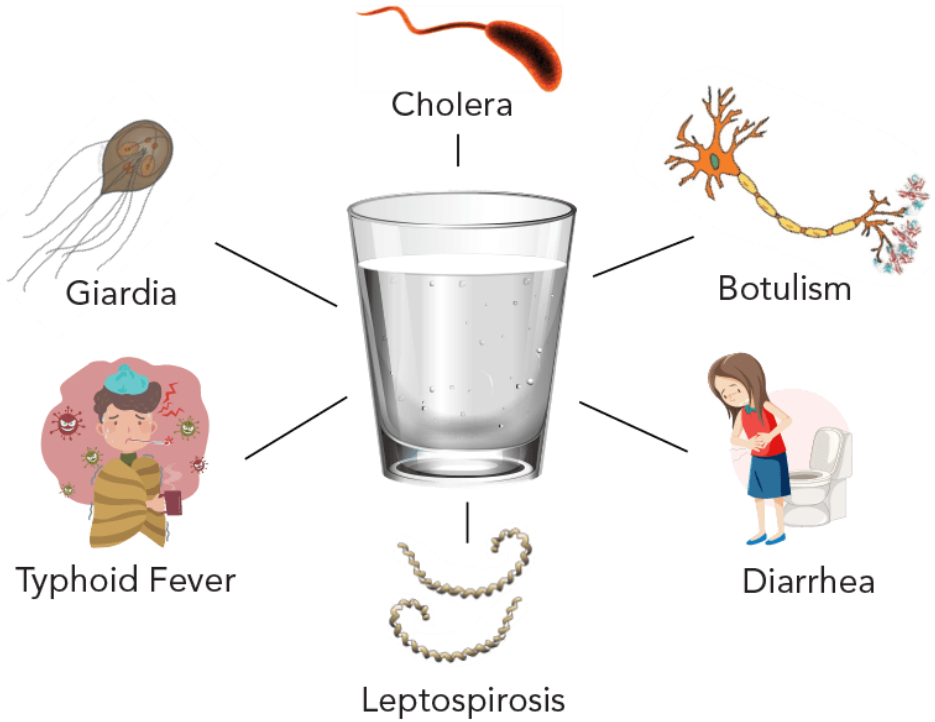
NOTES

1 In case of dispute, the method indicated by '*' shall be the referee method.

2 It is recommended that the acceptable limit is to be implemented. Values in excess of those mentioned under 'acceptable' render the water not suitable, but still may be tolerated in the absence of an alternative source but up to the limits indicated under 'permissible limit in the absence of alternate source' in col 4, above which the sources will have to be rejected.

Pathogens

- From the perspective of human use and consumption, the most important biological organisms in water are pathogens, those organisms capable of infecting or of transmitting diseases to, humans.
- These organisms are not native to aquatic systems and usually require an animal host for growth and reproduction.
- They can.. however, be transported by natural water systems, thus becoming a temporary member of the aquatic community.
- Many species of pathogens are able to survive in water and maintain their infectious capabilities for significant periods of time. These waterborne pathogens include species of bacteria, viruses, protozoa, and helminths (parasitic worms).



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: mfine, https://www.mfine.co/guides/waterborne-diseases-india

List of water borne diseases

Disease	Microbial Agent	Disease Symptoms	Chronic Effect
Campylobacteriosis	Bacterium (*Campylobacter jejuni*)	Fever, abdominal pain, diarrhea	Chronic sequelae, such as reactive arthritis and Guillain-Barré syndrome (GBS
Cholera	Bacterium (*Vibrio cholerae)*	Watery diarrhea, vomiting, occasional muscle cramps	significant decrease in the pertussis-toxin-catalysed ADP-ribosylation, prolactin secretion increased
Cryptosporidiosis	Protozoan (*Cryptosporidium parvum*	Diarrhea, abdominal discomfort	the small intestine is most commonly affected, Cryptosporidium infections could possibly affect other areas of the digestive tract or the respiratory tract.
Giardiasis	Protozoan (*Giardia lamblia*)	Diarrhea, abdominal discomfort	leak flux, malabsorptive and secretory components
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Amebiasis	Protozoan (*Entamoeba histolytica*)	Abdominal discomfort, fatigue, diarrhea, flatulence, weight loss	Colitis, Appendicitis, Peritonitis, Liver abscess, Lung abscess
Hepatitis	Virus (hepatitis A)	Fever, chills, abdominal discomfort, jaundice, dark urine	Numbness in extremities. Mental confusion / 'brain fog Dizziness & peripheral vision problems. Cognitive dysfunction Shortness of Breath Visual Changes, Female Problems (irregular menses, severe PMS)

Source: 1. Das, D. (2009). Waterborne diseases: linking Public Health and watershed data. Masters Theses, 235.

List of water borne diseases

Disease	Pathogen	Signs and Symptoms	Transmission	Diagnostic Tests	Antimicrobial Drugs
Strongyloidiasis	*Strongyloides stercoralis*	Often asymptomatic; cough (sometimes bloody), skin rash, abdominal pain, diarrhea; in immunosuppressed patients, may become disseminated, causing serious and potentially fatal complications	Soil-dwelling larvae penetrate the skin, usually bare feet	Microscopic observation of larvae in stool; serological testing for antigens	Ivermectin, albendazole
Tapeworms (taeniasis)	*Taenia solium, T. saginata, T. asiatica, Diphylloboth-rium latum*	Asymptomatic or mild GI distress; cysts in muscle, eye, or brain (cysticercosis); brain cysts can cause headaches, seizures, or death	Ingestion of raw or undercooked pork or beef from infected animal	Observation of worm segments or microscopic eggs in stool; CT or MRI to detect cysts	Praziquantel, niclosamide
Trichinosis	*Trichinella spiralis,* other *Trichinella* spp.	Diarrhea, constipation, abdominal pain, headache, cough, chills, light sensitivity, muscle pain, fever, conjunctivitis; in severe cases may affect motor coordination, breathing, heart function	Ingestion of raw or undercooked pork or other meat of infected animal	Observation of cysts in muscle biopsy, enzyme immunoassay	Albendazole, mebendazole
Whipworm (trichuriasis)	*Trichuris trichiura*	Abdominal pain, anemia, diarrhea (possibly bloody), rectal prolapse	Ingestion of eggs in fecally contaminated food	Microscopic observation of eggs in stool	Albendazole, mebendazole, ivermectin

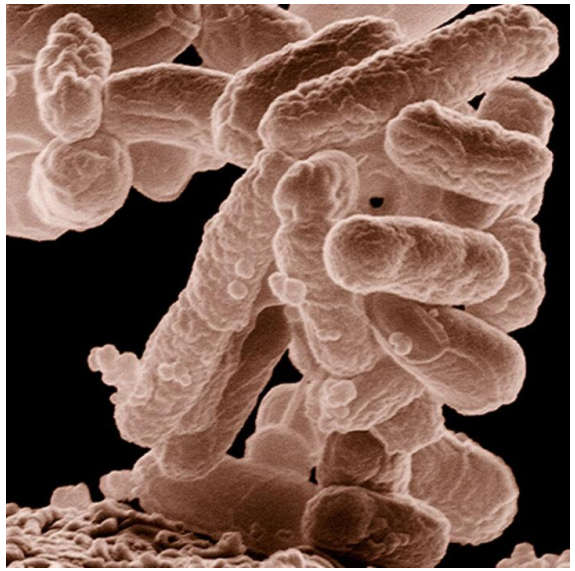
Source: 1. Helminthic Infections of the Gastrointestinal Tract, https://bio.libretexts.org/Bookshelves/Microbiology/Microbiology_%28OpenStax%29/24%3A_Digestive_System_Infections/24.06%3A_Helminthic_Infections_of_the_Gastrointestinal_Tract

Pathogens

- Most of the waterborne pathogens are introduced through fecal contamination of water. Thus, any organism native to the intestinal tract of humans and meeting the above criteria would be a good indicator organism. The organisms most nearly meeting these requirements belong to the fecal coliform group.

- Composed of several strains of bacteria, principal of which is ***Escherichia coli***, these organisms are found exclusively in the intestinal tract of warm-blooded animals and are excreted in large numbers with feces.
- Fecal coliform organisms are non-pathogen and are believed to have a longer survival time outside the animal body than do most pathogens.

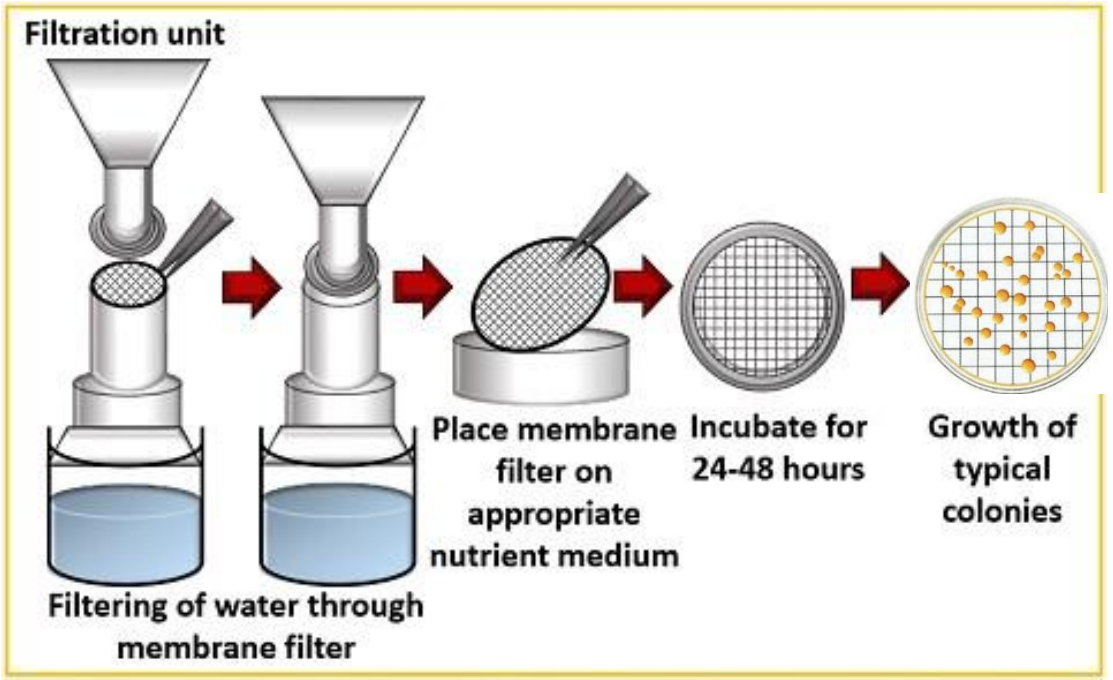
- Because the die-off rate of fecal coliforms is logarithmic, the number of surviving organisms may be an indication of the time lapse since contamination.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Escherichia coli O157:H7, https://food.unl.edu/article/escherichia-coli/ 3. Image: E. Coli Infection, https://weillcornell.org/news/e-coli-infection-how-to-stay-safe

Pathogens detection

- The **membrane-filter technique**, a technique popular with environmental engineers, gives a direct count of coliform bacteria.
- In this test, a portion of the sample is filtered through a membrane, the pores of which do not exceed 0.45 µm
- Bacteria are retained on the filter that is then placed on selective media to promote growth of coliform bacteria while inhibiting growth of other species.
- The membrane and media are incubated at the appropriate temperature for 24 h, allowing coliform bacteria to grow into visible colonies that are then counted.
- The results are reported in number of organisms per 100 mL of water.

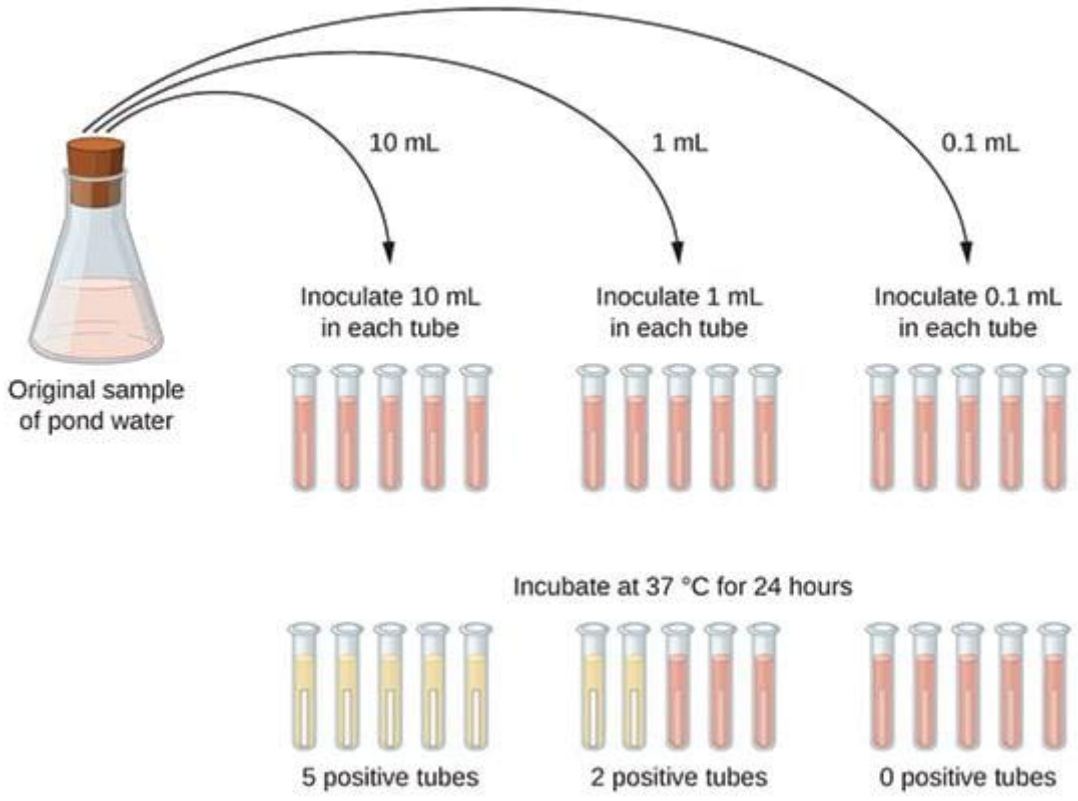


METHOD OF MEMBRANE FILTRATION

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: Membrane filtration method, https://biologyreader.com/membrane-filtration-method.html

Pathogens detection

- The **most probable number (MPN)** technique is a method for estimating the number of bacteria in a food or water sample.
- In this technique, replicate portions of the original sample are cultured to determine the presence or absence of microorganisms in each portion.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: MPN, https://microbenotes.com/water-quality-analysis-by-most-probable-number-mpn/ 3. Most Probable Number Technique, https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/most-probable-number-technique#:~:text=Introduction-,The%20most%20probable%20number%20(MPN)%20technique%20is%20a%20method%20for,of%20microorganisms%20in%20each%20portion.

Thank you

For any questions, please feel free to reach out to me.